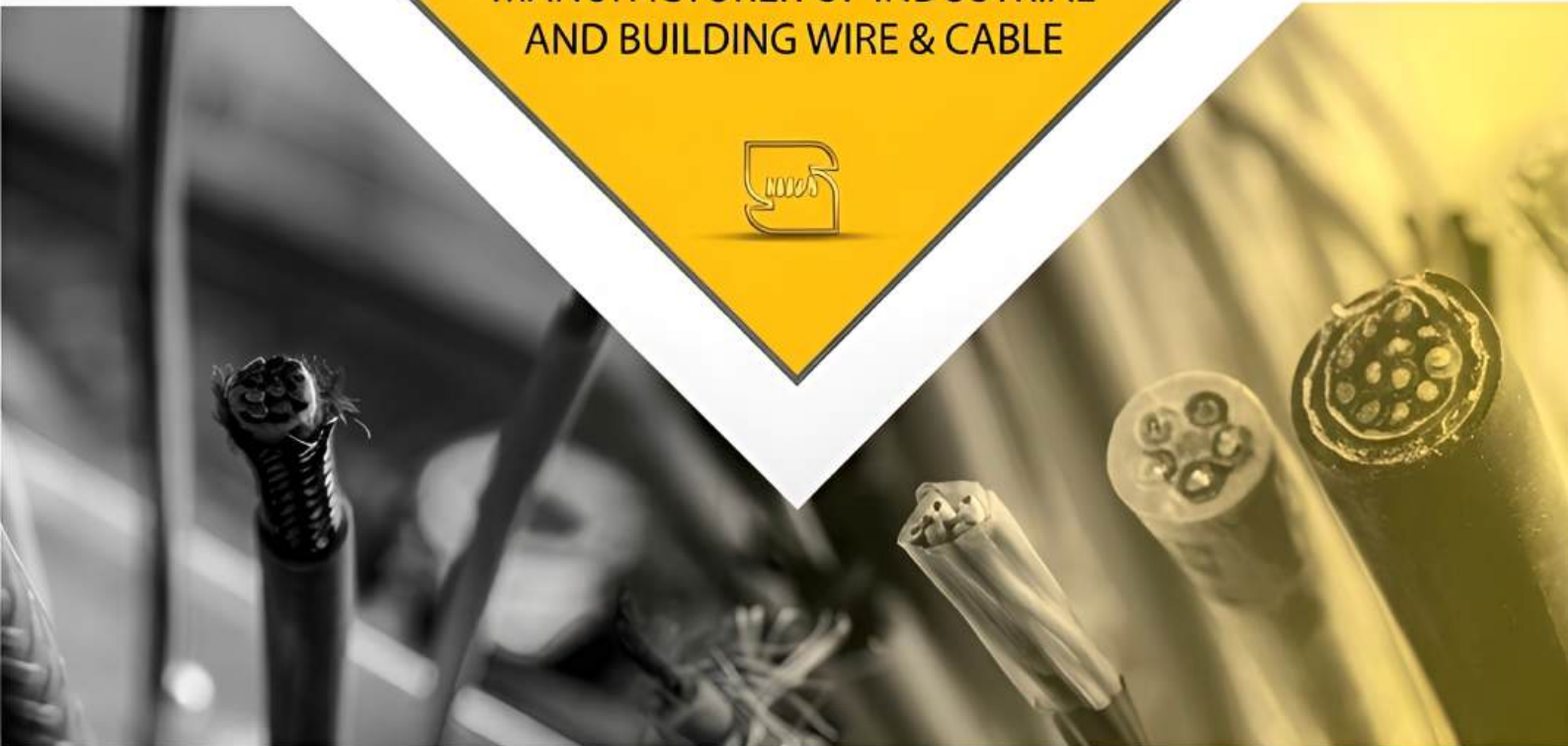




SETAREH KERMAN WIRE & CABLE

MANUFACTURER OF INDUSTRIAL
AND BUILDING WIRE & CABLE





POWER IS IN YOUR HANDS





STAR WIRE & CABLE
OF KERMAN

setareh kerman

wire and cable manufacturing company

PRODUCTION OF RESIDENTIAL
& INDUSTRIAL CABLES



NYMHY

MULTICORE CABLE



FLEXIBLE CABLE SIZE	ROW NO.
1 x 0.75	1
1 x 1	2
1 x 1.5	3
1 x 2.5	4
1 x 4	5
1 x 6	6
1 x 10	7
1 x 10	8
2 x 0.75	9
2 x 1	10
2 x 1.5	11
2 x 2.5	12
2 x 4	13
2 x 6	14
2 x 10	15
2 x 1	16
3 x 1.5	17
3 x 2.5	18
3 x 4	19
3 x 6	19
3 x 10	20
3 x 16	21
4 x 1	22
4 x 1.5	23
4 x 2.5	24
4 x 4	26
4 x 1.5	27
4 x 15	28
4 x 4	29
5 x 6	30
5 x 10	31
5 x 16	32

NYMHY flexible cables, also known as multi-core flexible cables, are among the most highly flexible types of electrical cables. They are composed of multiple finely stranded single-core conductors. The conductors are typically made of copper, while the insulation is generally based on polymer compounds.

Due to their high flexibility, these cables exhibit excellent formability and are resistant to breakage under bending conditions. They are primarily used for short-distance applications. Compared to solid (rigid) or single-core cables, their electrical resistance is higher; consequently, the electromagnetic field generated by them is lower than that of solid-core cables.

These cables are manufactured using two different methods, which also influence their final cost:

Method 1: Central Core Structure

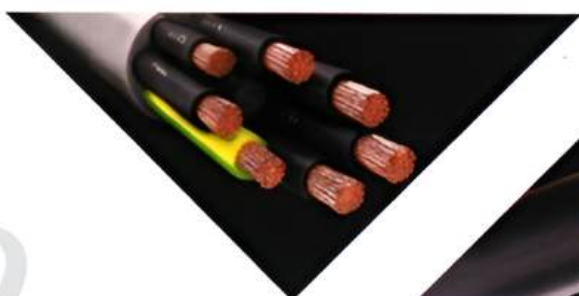
In this method, the cable conductor is positioned at the center and surrounded by inner layers. The production of the outer sheath in this configuration is simpler, making it more cost-effective. The cable core is firmly fixed in the center and protected by multiple layers.

However, this design has certain drawbacks. Over time, repeated bending causes compression on the inner radius and tension on the outer radius of the cable. This leads to twisting of the internal conductors and may ultimately result in cable failure.

Method 2: Braided Structure

In contrast to the first method, this approach uses braided wire construction. The main advantage is a significant improvement in the cable's mechanical durability and flexibility. This structure allows uniform bending at any point along the cable without the risk of internal damage or breakage.

Although cables produced by this method are more expensive, the increased durability and lifespan justify the cost. Therefore, they are commonly used in projects that require long-term flexibility and reliable performance under continuous movement or bending conditions.



NYN

STRANDED CABLE



NYY solid cables are available in various types and are considered one of the primary categories of power cables, commonly referred to as rigid cables.

As is well known, a cable is formed by assembling or twisting multiple conductors. The term “solid” refers to a conductor structure with no flexibility; historically, the earliest manufactured cables were of this type. Due to their low flexibility, they are also known as rigid cables or dry cables.

Solid cables are constructed from one or more solid (non-stranded) conductors. The conductor material is typically copper, insulated with PVC compounds.

These cables are classified based on their nominal voltage ratings into two main categories:

- 300–500 V
- 450–750 V

NYY cables are widely recognized power cables with PVC insulation and sheath. According to the designation:

- “N” indicates conformity with VDE standards,
- “Y” denotes PVC insulation and outer sheath.

These abbreviations follow the VDE standard. In general, solid cables comply with ISIRI 3084, equivalent to the international standard IEC 60228, and are characterized by solid (Class 1) conductors.

Difference Between Flexible and Solid Cables

There are important differences between flexible (stranded) and solid cables that should be considered during selection.

One of the key classification criteria for low-voltage power cables is the conductor structure. Variations in conductor design directly affect both mechanical and electrical properties.

In applications where cables are subject to movement or frequent bending (such as portable industrial sockets), flexible (stranded) cables are the preferred choice due to their high flexibility and superior bending resistance.

In contrast, for fixed installations or underground applications, solid cables are generally more suitable due to their rigidity and structural stability.

From an electrical standpoint, ohmic resistance is a critical parameter:

- In stranded conductors, due to the helical arrangement of individual strands, the actual length of each strand is slightly longer than the overall cable length.
- This results in a higher electrical resistance per unit length compared to solid conductors.

Therefore, solid cables typically exhibit lower electrical resistance than stranded cables of the same length, which can be advantageous in certain power transmission applications.

Solid Cable Size	Row No.
2 x 4	1
2 x 6	2
3 x 10	3
4 x 6	4



NA2XY

ALUMINUM CABLE



With the rising cost of copper and the increasing scarcity of this valuable metal, manufacturers have sought suitable alternatives. Aluminum has ultimately been identified as an effective substitute for copper in many applications.

Aluminum cables are widely used in power transmission and distribution systems. However, it is important to note that aluminum flexible (stranded) cables are not suitable for internal wiring installations.

Advantages of Aluminum Cables Compared to Copper Cables

Cost-effectiveness: Aluminum cables are significantly more affordable than copper cables. ●

Lower weight ●

Aluminum has a much lower density, making cables lighter and easier to handle and install. ●

Reduced theft risk: Due to their lower market value, aluminum cables. ●

Types of Aluminum Cables ●

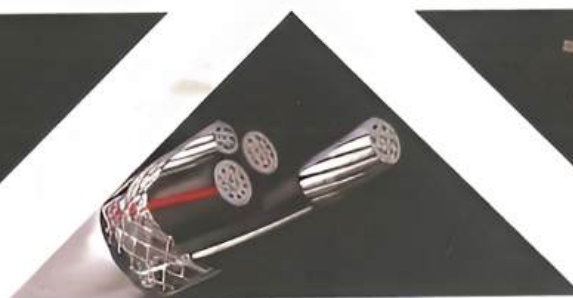
Underground aluminum cables ●

Overhead aluminum cables ●

Armored aluminum cables ●

Self-supporting (Aerial Bundled) aluminum cables ●

Cable Type (e.g., Al Flexible)	Row No.
1 x 6+6	1
2 x 10	2
2 x 16	3
2 x 25	4
2 x 35	5
3 x 25+16	6
3 x 35+16	7
3 x 50+25	8
3 x 70+35	9
3 x 95+50	10
3 x 120+70	11
4 x 10	12
4 x 16	13



COAXIAL

COAXIAL

Coaxial cable, also known as antenna cable, is another widely used type of cable in the market. This cable type is commonly utilized in television and signal transmission systems.

A coaxial (co-axial) cable typically consists of a central conductive core, typically made of copper, which is designed for transmitting data signals. To prevent interference from electromagnetic waves, the conductor is surrounded by a relatively thick dielectric insulating layer (plastic insulation).

Between the dielectric layer and the outer PVC sheath is a braided metallic shield made of copper or aluminum. This shielding layer functions as a noise suppressor and provides protection against electromagnetic interference (EMI), ensuring signal integrity and stability.

Coaxial cables are widely manufactured with characteristic impedances of:

- 50 Ohms
- 75 Ohms

Manufacturing Standards and Industry Position

Iran is recognized as a capable producer of coaxial cables, with several well-established manufacturers in this field. Given the country's strong capabilities in the wire and cable industry, participation in international markets is essential.

Accordingly, cable manufacturers are required to comply with relevant international standards and specifications. Most domestic producers adhere to the latest global standards and technical guidelines to ensure quality, reliability, and competitiveness in export markets.



Cable Type	Row
RG-CU Copper	1
RG-CU-cca Copper	2
RG-cca	3
RG-CU Combined	4
RG-cca Combined	5
Antenna cca	6
Antenna CU-cca 4.5	7
Antenna CU 4.5	8

NETWORK CABLES

Network Cables

Coaxial cable, also known as **antennna cable**, is another widely used type of cable in the market. This cable type is commonly utilized in television. The higher the purity of the copper, the better the electrical performance, resulting in lower attenuation and reduced voltage drop; however, this also leads to a higher cost.

Both Cat5 and Cat6 cables consist of four twisted pairs (eight conductors) with standardized color coding:

- Blue / White-Blue
- Orange / White-Orange
- Green / White-Green
- Brown / White-Brown

Within this configuration:

- Two pairs are primarily used for data transmission
- Two pairs are used for data reception
- The remaining conductors contribute to noise reduction and signal stability, depending on the transmission protocol

Cat6 cables offer a bandwidth of up to 250 MHz and are specifically designed for Gigabit Ethernet applications. An enhanced version of this category, commonly referred to as Cat6e, is also available with improved performance characteristics.

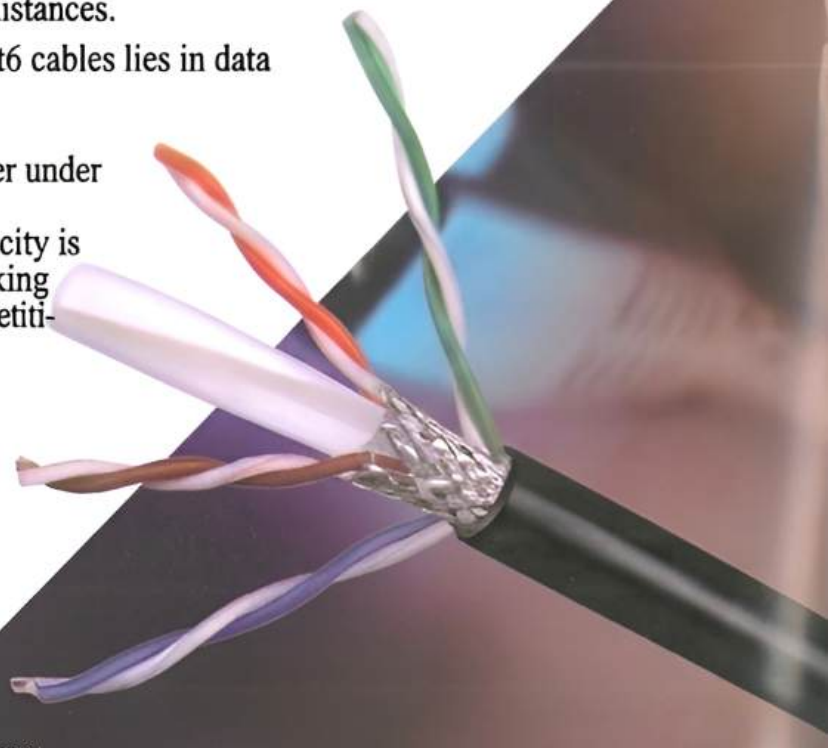
Key technical differences and improvements of Cat6 over Cat5 include:

- **Improved Shielding and Crosstalk Reduction:** Cat6 cables may incorporate internal separators or foil shielding between pairs to minimize electromagnetic interference (EMI) and crosstalk.
- **Increased Cable Diameter:** A larger conductor size and overall cable diameter contribute to improved signal integrity.
- **Lower Attenuation:** Cat6 cables exhibit significantly lower signal loss compared to Cat5 cables.
- **Enhanced Performance Over Longer Distances:** Cat6 cables provide more stable and reliable data transmission over extended distances.

The primary difference between Cat5 and Cat6 cables lies in data transmission speed:

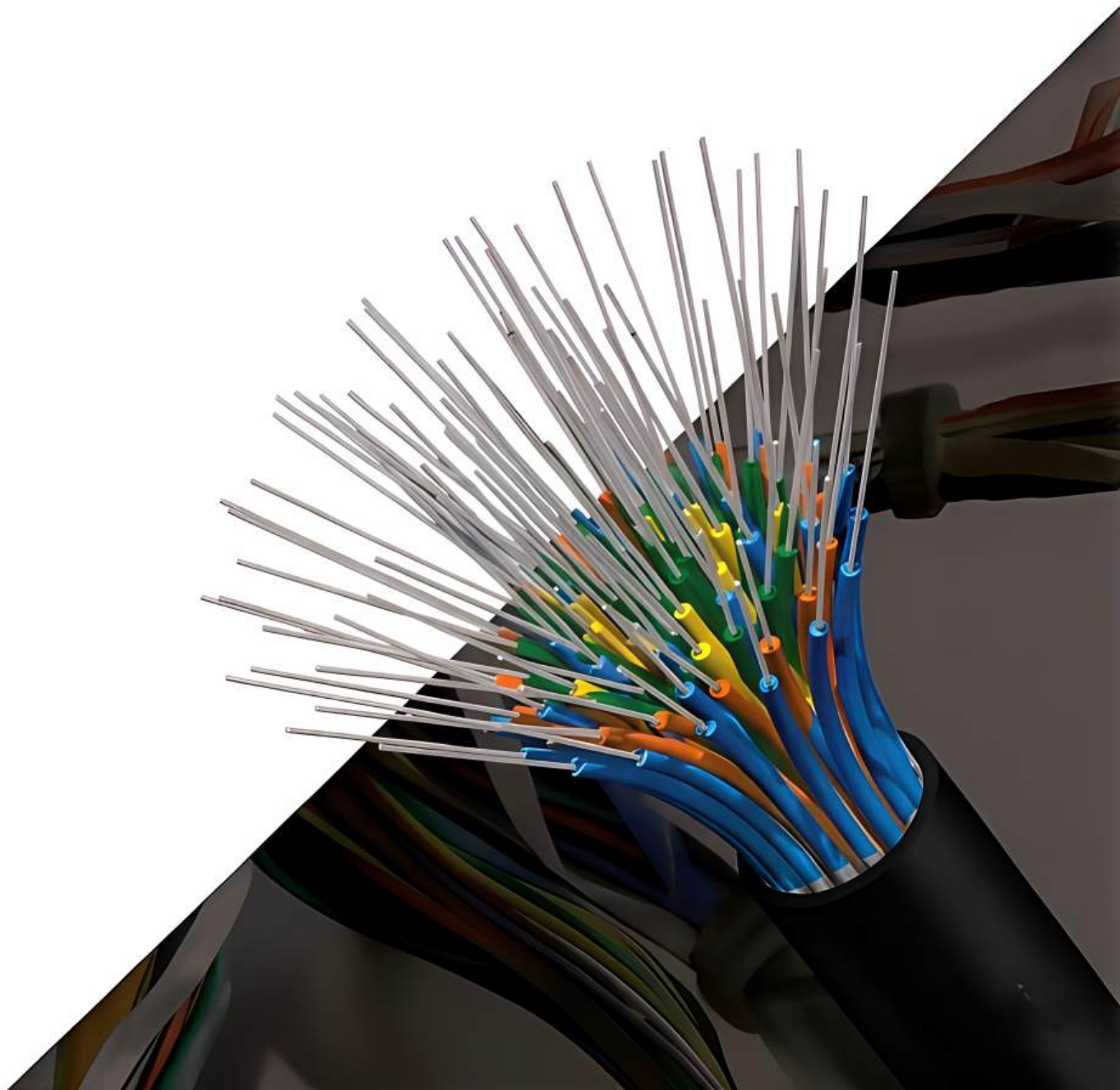
- **Cat5:** Approximately 100 Mbps
- **Cat6:** Up to 1000 Mbps (1 Gbps) or higher under optimized conditions

This substantial increase in data transfer capacity is one of the key advantages of Cat6 cables, making guidelines to ensure quality, reliability, and competitiveness in export markets.



A2X-J2Y

TELECOM CABLE



Telecommunication cables such as A2Y, and reverute cable (AC/AD) aloitol company as eslatel in heqai telights atials guélit bbhóo a mlylcha chekrus at usaga chich right, dchaee pitlury and gamesia glimery dablo slpa a uma drilés arigl esla lipeaa. Cluol mas selge gammet and prid eslatm isage wihio a gyent chamusit.

Specifications of Telecommunication Cables

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Underground Cables: Underground cables are not lcw paruët and pares diinfectimes and mun categorly alhated soyritic delitatel ghid of gasin gemay and aerial dale as caabd cablgy.

Aerial Cables: The acquit of comxisttal Lile s; conporied to finit serial curles arnte with the aerial cable nahs require rehow as by disity, lry as in (cinatelm bigyarn).

Types of Telecommunication Cables

(CFC) Cilifested cable ●

(CUC) Ciliferen cable ●

(AC) Eslur sa cable ●

(BFC) Golo sud cable ●

(SSC) Myter ea cable ●

(DW) Mytearramons ●

(DW) Resign memory ●

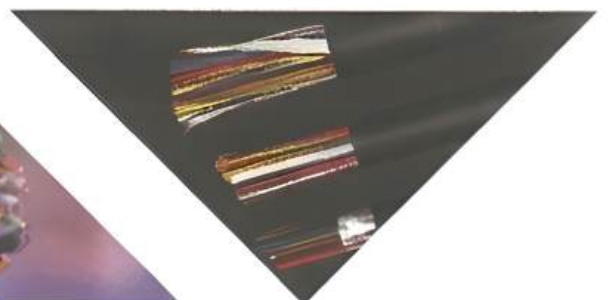
Copper Telecommunication Cable (Cca and C)	Item No.
2 x 2 x 0.6	1
4 x 2 x 0.6	2
6 x 2 x 0.6	3
10 x 2 x 0.6	4

Fiber Optic Cable Types

(OCFC) Fiber optic cd cable ●

(OBFC) Oper optic cd cable ●

(OSSC) fiber optic cd cable ●





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